

Yinzhe Zhou

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EDUCATION

Johns Hopkins University

Master of Science in Electrical and Computer Engineering

Baltimore, MD, USA

Aug. 2025 – Present

- **GPA:** 3.73/4.0
- **Relevant Coursework:** FPGA Synthesis Lab, CAD Design of Digital VLSI Systems, Advanced Topics in Microsystem Fabrication, Neural Implants and Interfaces, Mixed-Mode VLSI Systems.

University of Macau

Bachelor of Science in Electrical and Computer Engineering

Macao SAR

Aug. 2021 – May 2025

City University of Hong Kong

Exchange Student, Department of Electrical Engineering

Hong Kong SAR

Jan. 2023 – Aug. 2023

MANUSCRIPTS

Learning Haptic Active Compliance for Force-Aware Dexterous Manipulation

Naisheng Ye, **Yinzhe Zhou** (co-first author), Junkai Zhao, Yuhang Lu, Checheng Yu, Zhenjie Yang, Pengwei Wang, and Hongyang Li. *Unpublished manuscript, 2026.*

RESEARCH INTERNSHIPS

Beijing Academy of Artificial Intelligence (BAAI)

Research Intern

Beijing, China

Jun. 2026 – Sep. 2026

HACo: Haptic Active Compliance for Dexterous Manipulation

- Developed compliance-regulated teleoperation for dual UR5 arms with Sharpa dexterous hands, combining Cartesian admittance and fingertip force regulation; collected **500 demonstrations** across five tasks with synchronized vision, robot state, tactile, and joint-torque observations.
- Developed a kinematics-aligned haptic encoder fusing ten fingertip tactile streams with 44 hand-joint torques, and gated cross-attention to condition a pretrained vision-language-action model on contact and load feedback.
- Trained the policy with conditional flow matching on compliant-action targets and auxiliary compliant-intent supervision; integrated real-time action chunking for asynchronous closed-loop execution.
- Conducted real-robot comparisons and component ablations across **five force-sensitive tasks**, with 20 trials per task-method pair; HACo achieved **83% mean success**, versus **35%** for the strongest evaluated baseline (**+48 percentage points**).

Microsoft Research Asia (MSRA)

Research Intern

Beijing, China

Jun. 2024 – Sep. 2024

Fine-Tuning Mixtral 8x7B on Biochemical Molecular Data

- Constructed and preprocessed a PubChem-based dataset for molecular property prediction tasks.
- Implemented KNN retrieval over SMILES representations to provide similar-molecule context for in-context learning.
- Fine-tuned Mixtral 8x7B and refined the evaluation protocol, improving test accuracy from **67% to 77%**.
- **Tech:** Python, Git, VS Code.

RESEARCH EXPERIENCE

Johns Hopkins University

Graduate Researcher

Baltimore, MD, USA

Feb. 2026 – Present

Scalable Shear-Force Sensing for Tactile Perception and Robotic Grasping

- Contributed to a scalable piezoresistive shear-force sensing system spanning multiple sensor form factors and integration onto a UMI handheld gripper for robot data collection.
- Developed embedded firmware and data-acquisition routines for real-time shear-field readout across different sensor-array configurations.
- Designed and ran sensor characterization and robotic grasping experiments to evaluate shear-force features under varied contact conditions.

Johns Hopkins University × TU Darmstadt
Graduate Researcher

Baltimore, MD, USA
Nov. 2025 – Present

Event-Based Visuo-Tactile Sensing for Robot Imitation Learning

- Building a multimodal imitation-learning pipeline that fuses RGB, tactile, and event streams for a UR robot-mounted gripper, targeting contact, slip, and fast-motion cues missed by RGB-only policies.
- Implemented a synchronized multi-rate data pipeline from handheld demonstrations into observation–action samples, aligning RGB at 30 Hz, tactile/event streams at 120 Hz, and 3D pose at 75 Hz.
- Trained and deployed **Diffusion Policy** and **Flow Matching** algorithms on real-world tasks.
- Developing an **event-conditioned Flow Matching** policy with a lightweight 3D-CNN event encoder for high-frequency tactile/event observations.

University of Macau

Final Year Project — Faculty Dean's Honors List

Macao SAR
Sep. 2024 – May 2025

Singular Spectrum Analysis for Biosignal Denoising and Feature Extraction

- Designed a two-level Singular Spectrum Analysis framework combining component correlation and waveform reconstruction for ECG and EEG signal analysis.
- Localized ECG PQRST features and removed EEG artifacts while preserving neural rhythms across delta, theta, alpha, and beta bands.
- Validated the method on synthetic and real recordings, recovering physiological patterns such as eyes-open vs. eyes-closed alpha rhythm differences.
- **Tech:** Python, NumPy, SciPy, Matplotlib, NeuroKit2.

City University of Hong Kong

Research Assistant

Hong Kong SAR
May 2023 – Aug. 2023

Neural Decoding for Implantable Brain–Machine Interfaces with ANN and SNN

- Preprocessed raw primate neural recordings into supervised datasets with movement coordinates, finger positions, and spike trains.
- Designed and benchmarked ANN and SNN decoders for neural decoding, comparing accuracy–efficiency trade-offs for implantable BMI scenarios.
- **Tech:** Python, MATLAB.

University of Macau

Research Assistant

Macao SAR
May 2022 – Jul. 2022

FPGA-Based Hardware Time-Triggered System for Biomedical Engineering

- Implemented FPGA-based signal conditioning and filtering pipelines in Quartus for biomedical data acquisition.
- Refined hardware architecture and I/O interfaces to support deterministic time-triggered behavior.
- **Tech:** Quartus, MATLAB.

SELECTED HONORS & AWARDS

1st Runner-Up, IEEE Student Branch Project Competition 2025.

Akrostar Academic Prize, University of Macau, 2024/2025.

Dean's Honor List, University of Macau, all semesters.

Academic Excellence Outgoing Exchange Scholarship, City University of Hong Kong, 2023.

Top 10%, Huawei ICT Competition (Preliminary), Macau, 2021–2022.

TECHNICAL SKILLS

Programming: Python, MATLAB, C/C++, Verilog, VHDL.

ML / Robotics: PyTorch, ROS 2, Diffusion Policy, Flow Matching, multimodal data synchronization, robot imitation learning.

Hardware / EDA: FPGA development, embedded firmware, signal acquisition, Vivado, Quartus, Cadence, 3D CAD.

Research Domains: Tactile sensing, biosignal processing, brain–machine interfaces, LLM fine-tuning, digital circuit design.